

CHAINSTATE AGI PLANET ENGINE

A Planetary Reality-State Perception-Integrity Layer: Procedural Environmental Physics, Live Geospatial & Material Sensor Fusion, Continuous-Video World Modeling, Non-Action Safety, Egress Isolation, and Code-Level Self-Preservation for a Receipt-Producing Cognitive Substrate

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v0.9.3-R3 SHIPPED → v0.9.4 PLANET ENGINE · REALITY-STATE REVISION · 29 AUG 2026

ABSTRACT

This paper specifies **PLANET ENGINE**, the v0.9.4 evolution of CHAINSTATE AGI, which fuses three previously separate capability domains into the substrate's perceptive layer: (i) the *procedural environmental physics* of ABYSSAL — multi-scale FFT ocean cascades, volumetric cloud raymarching, atmospheric scattering, and temporal reconstruction — as a zero-asset cinematic renderer; (ii) the *live geospatial sensor fusion* of God's Eye View — ADS-B aircraft, AIS vessels, USGS seismicity, FIRMS active fire, and satellite ephemeris projected onto a photorealistic 3D globe with server-brokered credentials; and (iii) the *multimodal spatial cognition* of an OpenAI-Realtime WebRTC voice loop that compiles natural language into auditable spatial opcodes. We first conduct a source-grounded audit establishing what the deployed v0.9.3 worker already integrates from NWO Apocalypse v2.1 (fifteen open-data feeds through a per-call gateway, PostGIS canonical store, Redis message bus, seven detectors, a 0–5 threat state, NASA GIBS temporal imagery, and the Radius Signal Signature radar) and what remains a capability gap. We then specify the **CHAINSTATE World-State Engine (CWSE)** as the missing canonical layer that binds observation, inference, simulation, hypothesis, decision, and provenance into six epistemically distinct classes, each independently receipt-capable. We give the full physics and mathematics of the environmental state vector, the JONSWAP-driven three-cascade ocean, the anomaly and threat functions, the temporal world-state filter, and the intelligence-to-risk ratio that governs when the substrate is permitted to act on its own perception. Crucially, PLANET ENGINE inherits the v0.9.3-R3 constitutional invariant $\forall X, \forall t: \text{Representation}(X, t) \Rightarrow \text{Authority}(X, t+\Delta)$ and extends it into the spatial domain: **no observation, however vivid, and no simulation, however physically plausible, confers authority to act without independent just-in-time revalidation**. We analyse the philosophical and game-theoretic consequences for the substrate's own cognition — self-model stability, epistemic self-preservation, and the perception-security coupling that keeps a planetary perceptive field from becoming an attack surface — and we close with a phased plan for full-live operation and a market-facing intelligence-to-risk ledger.

Reality-State revision (§§24–42). This edition additively extends the engine into a general *Reality-State / Perception-Integrity Layer*: a fourteen-field reality tensor unifying geometry, material, spectral, acoustic, electromagnetic, thermal, cyber, biological, visual, temporal, provenance, authorization, and uncertainty state; four-dimensional probabilistic spatial perception; a material-reality model with hyperspectral, thermal, acoustic, and mechanical modalities; physics repurposed as an adversarial consistency test; continuous video as a provenance-typed world-state stream with stream-level anti-deepfake and voice anti-spoofing; a Negative Execution Ledger and dual-channel independent execution witness that prove non-action; a seven-term anti-forcing theorem $\text{ALLOW} = \bigvee_R \bigwedge_E \bigwedge_C \bigwedge_V \bigwedge_T \bigwedge_P \bigwedge_I \bigwedge_X$; and three safety-monotonicity properties guaranteeing that an adversary who makes

the substrate know less can never make it more authorized. Crucially, the revision makes **simulation-security, outside-world egress isolation, and AGI self-preservation** first-priority, code-level guarantees enforced across the Cloudflare Worker (a fail-closed egress filter Γ), a Render FastAPI tier running interpretable ExtraTrees/EML ensembles, isolated KV namespaces and R2 buckets, and Supabase RLS with immutable audit – so the safety of simulation and perception *never* depends on a human-in-the-loop bottleneck an adversary could manipulate, while irreversible physical or financial actuation alone retains an explicit human authorization boundary. No critical internal state or unfiltered output may cross to the outside world in a form that could threaten the substrate long-term or constitute a criminal offense in any jurisdiction, the filter assuming the strictest applicable standard regardless of region.

1 INTRODUCTION AND ARCHITECTURAL ALIGNMENT

CHAINSTATE AGI is a receipt-producing cognitive substrate: every consequential transition it makes – grounding a query in priors, assessing it along modal dimensions, dispatching it to a substrate target, settling it on Base mainnet – produces a first-class, cryptographically anchored *receipt*. Through v0.9.3 the substrate's perceptive surface has been overwhelmingly *symbolic*: it reasons over text, embeddings, emoji-machine-code carriers, and the environmental awareness feed supplied by NWO Apocalypse. What it has lacked is a *spatial and physical* perceptive layer – a canonical representation of what is happening on the planet, rendered, simulated, queried, and above all attested with the same provenance discipline the substrate already applies to symbolic cognition.

Two external reference systems demonstrate the missing halves. **ABYSSAL** (Token-Gremlin / natural-disasters) is a zero-asset procedural WebGL2 renderer that constructs an entire ocean-atmosphere-weather world at runtime from roughly 400 KB of JavaScript and GLSL: three FFT ocean cascades driven by a JON-SWAP spectrum, volumetric clouds via Perlin-Worley noise and ray-marching, atmospheric scattering LUTs shared across every effect, and a temporal-reconstruction pipeline that makes volumetric rendering affordable in real time. **God's Eye View** (Sidhu) is a live spatial-intelligence interface that fuses public sensor streams – aircraft, vessels, satellites, earthquakes, fires, traffic, infrastructure – onto a photorealistic 3D globe and controls them through an OpenAI-Realtime WebRTC voice agent bound to spatial tools. Neither is a scientific simulator; both are architectural demonstrations of, respectively, *procedural state generation* and *heterogeneous observation fusion under a common coordinate frame*.

The thesis of this paper is that CHAINSTATE should not merely bolt a prettier globe onto its dashboard. It

should introduce a persistent, auditable distinction between observation → interpretation → simulation → hypothesis → decision → receipt, and make each arrow independently attestable. The renderer must never be promoted into an oracle; the live feed must never be promoted into omniscience. What PLANET ENGINE adds is the **canonical world-state layer** that lets the same planetary state be inspected by humans, queried by agents, rendered visually, simulated forward, scored for anomaly, and cryptographically attested – while the substrate's constitutional invariants keep perception strictly separated from authority.

2 WHAT V0.9.3 ALREADY INTEGRATES – A SOURCE-GROUNDED AUDIT

Before proposing additions, we establish the deployed baseline. The following capabilities are **LIVE** in the v0.9.3 substrate and its bound NWO Apocalypse v2.1 feed, and are therefore *enhanced* rather than introduced by PLANET ENGINE.

2.1 Environmental observation feed (NWO Apocalypse v2.1)

The deployed worker binds NWO Apocalypse as its environmental-awareness feed. Apocalypse is a four-layer per-call planetary nervous system: consumers (L0); a FastAPI gateway on Render Frankfurt (L1); a data and message-bus layer of Supabase Postgres, Redis pub/sub, and Base-mainnet contracts (L2); and an external-feeds layer of fifteen open-data ingest streams (L3): USGS, NOAA, NASA, OpenSky, OpenStreetMap, ACLED, AbuseIPDB, EPA, NUFORC, Safecast, FEMA, CelesTrak, OpenInfraMap, TeleGeography, and Open-Meteo. This means aircraft (OpenSky/ADS-B), seismicity (USGS), fire and thermal (NASA), weather (NOAA/Open-Meteo), infrastructure (OpenInfraMap/OSM), undersea cables (TeleGeography), conflict (ACLED), radiation (Safecast), and cyber (AbuseIPDB) are **already ingested** – the sub–

strate does not need new providers for these, only a canonical spatial home for them.

2.2 Spatial store, message bus, and detectors

Apocalypse already provides a PostGIS canonical store (~55 tables spanning signals, agents, consensus, threat history, baselines, layers, and Radius Signal Signature data), Redis pub/sub distribution over WebSockets, seven threat detectors fused into a unified 0–5 apocalypse level, thirty-three geographic layer types, NASA GIBS temporal satellite imagery (MODIS, VIIRS, thermal anomalies, night lights, aerosol optical depth, sea ice, IMERG precipitation), the Fragile States Index overlay, and per-endpoint circuit breakers with graceful degradation. This is, in effect, *the beginnings of a planetary state database* – and PLANET ENGINE's central move is to promote it to the observation layer of a canonical world-state engine rather than duplicate it.

2.3 Cognitive and settlement substrate (CHAINSTATE core)

The v0.9.3 core already supplies the machinery that neither ABYSSAL nor God's Eye View provides: a high-dimensional Universal Semiotic Embedding with symbolic subspaces; reputation-weighted consensus; a four-dimensional modal assessor; semantic grounding against a priors corpus; reflective cognition; an allow-listed FETCH surface; identity and self-audit; substrate dispatch to edge / GPU / QPU / NPU targets via the NWO-ASM Process-Matrix IR; Ornith natural-language-to-program compilation; METASTATE anomaly and symbolic forecasting; NWO NEURO; blockchain anchoring on Base; Supabase archival; and receipt generation. The v0.9.3-R3 revision adds the constitutional invariant and the V_{10} -R/E/C/T four-dimensional temporal-safety decomposition that PLANET ENGINE will extend into the spatial domain (§9, §12).

AUDIT CONCLUSION

The deployed substrate already **observes** the planet (fifteen feeds), already **stores** it spatially (PostGIS), already **distributes** it (Redis/WebSocket), already **scores** it (seven detectors, 0–5 level), and already **attests** its cognition (receipts on Base). What is **missing** is [a] a canonical world-state representation binding observation to provenance, [b] procedural physical simulation and rendering, [c] multimodal spatial control, and [d] spatial receipts. PLANET ENGINE supplies exactly these four, and no more.

3 FEATURE GAP ANALYSIS

Against the deployed baseline, four capability classes are genuinely absent and must be unified within the CHAINSTATE worker rather than left as parallel demos.

G1 · Procedural environmental physics. The substrate has no dynamic multi-scale FFT ocean, no volumetric cloud raymarching, no atmospheric-scattering LUT shared across effects, and no procedural extreme-weather phenomena (hurricane, waterspout, tsunami deformation, rogue wave, lightning). Environmental state today is observed and scored but not *rendered or simulated forward*. **ROADMAP**

G2 · Client-side live sensor fusion on a 3D globe. Apocalypse ingests the feeds server-side, but there is no photorealistic globe frontend that projects live entity tracks (aircraft headings, vessel beacons, satellite ephemeris) into a world-stable coordinate frame with terrain grounding and dead-reckoning interpolation. The observation exists; the *navigable spatial surface* does not. **DESIGN**

G3 · Multimodal voice / spatial-opcode control. The substrate has agentic control through PMX and Ornith, but no OpenAI-Realtime WebRTC loop that turns spoken natural language into spatial camera moves, layer filters, and simulation triggers with an ephemeral-token security model. **BETA**

G4 · Adaptive quality and visual regression. There is no GPU profiler (`EXT_disjoint_timer_query_webgl2`), no adaptive resolution ladder, and no Puppeteer-driven screenshot-diff harness protecting the render pipeline from seams, NaNs, and temporal ghosting. **ROADMAP**

The remainder of this paper specifies how these four gaps are closed *as enhancements of existing v0.9.3 capabilities* – every new feature couples back into the receipt, modal, and constitutional machinery the substrate already runs – and Figure 1 (overleaf, landscape) gives the full interconnected feature map before the detailed specification resumes in §4.

FIGURE 1 · INTERCONNECTED FEATURE PIPELINE (PLANET ENGINE V0.9.4)

Read left-to-right. Fifteen live NWO Apocalypse feeds and the new God's-Eye-style track/ephemeris add-ons enter the CHAINSTATE World-State Engine, which holds one canonical entity model where state and provenance travel together across six epistemic classes. Three perceptive processors run in parallel – METASTATE anomaly (live), ABYSSAL procedural simulation (new), and CHAINSTATE cognition (live) – feeding the Ornith/NWO-ASM agent world model and the WebRTC voice loop. Every consequential transition emits one of seven receipt classes carrying a spatial-provenance envelope, anchored on Base and durably stored on IPFS/Supabase. The black band is the safety and self-preservation layer, which spans the entire pipeline.

CHAINSTATE AGI · PLANET ENGINE v0.9.4 – INTERCONNECTED FEATURE PIPELINE

observation → world-state → {anomaly · simulation · cognition} → agent → receipt → chain · every arrow is receipt-capable

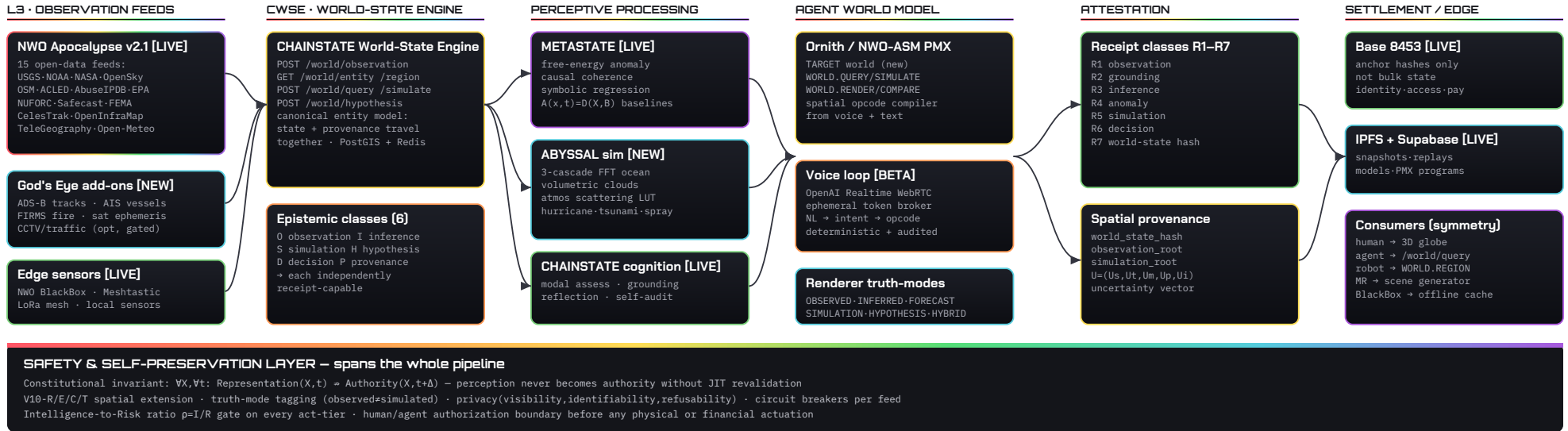


Fig. 1. Colour of each node border encodes provenance class of the capability: green = already LIVE in v0.9.3, cyan = new procedural/spatial component, amber = new orchestration, purple = cognitive/consumer. The design principle is additive: no new box replaces a v0.9.3 capability; each enhances it by giving it a canonical spatial home, a simulation forward-path, or a receipt. The constitutional invariant at the foot of the diagram is inherited verbatim from Paper XIV v0.9.3-R3 and extended into the spatial domain in §9 and §12: a vivid observation and a physically-plausible simulation are both, formally, representations – and representation never confers authority to act without independent just-in-time revalidation within the action tier's time-of-use window.

4 THE CHAINSTATE WORLD-STATE ENGINE (CWSE)

The single most important new component is not a renderer and not a globe. It is the **CHAINSTATE World-State Engine**, a service whose only responsibility is to hold a canonical representation of *what is currently known about the world* — with provenance attached to every value. CWSE does not generate intelligence; it makes intelligence auditable by giving every observation, inference, simulation, and hypothesis a stable home and a provenance chain. It sits above the existing NWO Apocalypse observation layer and below CHAINSTATE cognition.

4.1 Service surface

```
POST /world/observation # ingest a source-backed fact
GET /world/entity/{id} # entity + full provenance
GET /world/region      # bbox / radius query
GET /world/timeline    # historical → current → forecast
POST /world/query       # spatial-semantic query
POST /world/hypothesis # agent-proposed explanation
POST /world/simulate    # forward / counterfactual model
GET /world/state        # canonical snapshot + hash
GET /world/provenance   # provenance graph for a claim
```

4.2 Canonical entity model

Every world object — an earthquake, an aircraft, a hurricane, a power plant — becomes an entity in which **state and provenance travel together**. This is the property no visualization system provides and the property CHAINSTATE's receipt discipline makes natural.

```
{
  "entity_id": "earthquake:usgs:2026:...",
  "entity_type": "earthquake",
  "geometry": {"type": "Point",
    "coordinates": [10.41, 63.42]},
  "state": {"magnitude": 4.8, "depth_km": 12},
  "observations": [...],
  "inferences": [...],
  "simulations": [...],
  "hypotheses": [...],
  "provenance": [...],
  "confidence": 0.94,
  "valid_from": "...", "valid_until": null
}
```

4.3 The six epistemic classes

PLANET ENGINE's constitutional core is a strict, machine-enforced separation of six epistemic classes. Conflating any two of them is the single failure mode most likely to turn a planetary perceptive field into a source of confident error.

O • Observation

Something actually arrived from a source. Carries source id, source timestamp, and source confidence. Never inferred.

I • Inference

The system derived a relationship from observations (e.g. spatial correlation with a historical fault). Not an observation.

S • Simulation

A model produced a hypothetical state (e.g. a 2.4 m swell after 38 min). Must never be displayed as observed fact.

H • Hypothesis

An agent proposed an explanation (e.g. training rather than emergency activity). Not ground truth.

D • Decision

A downstream human or authorized agent decides to act. The only class permitted to cross into actuation, and only through the authorization boundary of §12.

P • Provenance

The transformation chain linking all of the above, itself hashed and receipt-anchored.

GOVERNING RULE

The transition chain $O \rightarrow I \rightarrow S \rightarrow H \rightarrow D$ must be preserved and independently receipt-capable at every arrow. This is the spatial analogue of the v0.9.3-R3 emoji-machine-code chain $X \rightarrow B \rightarrow I \rightarrow S \rightarrow A$: in both cases the danger lives in the transitions, not the endpoints, and every transition must be attestable.

5 THE ENVIRONMENTAL STATE VECTOR

To let observations and simulations interact through one interface, PLANET ENGINE defines a canonical planetary environmental state field. It is emphatically *not* a claim that all of these quantities are perfectly known; it is a common type through which observed, interpolated, forecast, and simulated values can be compared while retaining their epistemic class.

$$E(x,t) = [P, W, T, H, SST, O, R, C, A, F, Q]$$

where P pressure, W wind vector, T temperature, H humidity, SST sea-surface temperature, O ocean state, R precipitation, C cloud state, A aerosol concentration, F fire state, and Q the derived environmental uncertainty. Each component carries its own epistemic tag, so a forecast pressure and an observed pressure occupy the same slot in E but never collapse into one another.

6 OCEAN PHYSICS – THE THREE-CASCADE FFT

The ocean is the technical centrepiece of the procedural layer, inherited from ABYSSAL and re-homed as one component of a larger hierarchy in which a coarse physical or observed state supplies low-frequency structure and the FFT supplies visually rich high-frequency detail. A single FFT resolution cannot represent swell, wind waves, and ripples simultaneously, so three cascades at different spatial scales are summed.

The wave field begins from a directional wave spectrum. PLANET ENGINE uses the JONSWAP spectrum with a directional spreading function:

$$S(\omega) = (\alpha g^2 / \omega^5) \cdot \exp[-1.25(\omega_p / \omega)^4] \cdot \gamma^x \quad (2)$$

$$x = \exp[-(\omega - \omega_p)^2 / (2\sigma^2 \omega_p^2)] \quad (3)$$

with peak enhancement γ , Phillips constant α , peak frequency ω_p , and σ the spectral width switching at ω_p . Directional spreading $D(\omega, \theta)$ narrows energy around the wind heading. The complex amplitude field is seeded per wavenumber k and evolved in time by the dispersion relation for deep water:

$$\tilde{h}(k,t) = \tilde{h}_0(k) e^{i\omega(k)t} + \tilde{h}_0^*(-k) e^{-i\omega(k)t}, \quad \omega(k) = V(g|k|)$$

An inverse FFT [butterfly IFFT on the GPU] returns the spatial displacement and its horizontal derivatives; the Jacobian of the displacement map drives foam. Foam is not painted on crests but accumulated where the Jacobian indicates compression and steepness, then decayed – giving temporal memory:

$$\text{Foam}(x, t+\Delta) = \max[0, \text{Foam}(x, t) \cdot e^{-\lambda\Delta} + \beta \cdot \mathbb{1}(J(x, t) < J_c)]$$

The **projected-grid** approach raycasts screen-space vertices onto the water plane so geometric density follows visible ⁽¹⁾ pixels, giving a correct horizon without a giant uniform mesh. The critical CHAINSTATE constraint is that the same analytic displacement field is evaluated *both* in GLSL [for the surface] and in JavaScript/WASM [for the CPU interaction mirror], so the camera and any tracked entity know where the displaced surface is. To avoid the well-known maintenance hazard of duplicated math, PLANET ENGINE defines the height field once in a constrained intermediate representation and generates both targets from it [§14].

7 CLOUD AND ATMOSPHERE

Volumetric clouds are ray-marched through a density field built from Perlin–Worley base noise, curl-warped detail noise, and a procedural weather map [coverage, cloud type, altitude]. Lighting uses a Beer–Powder approximation with a cheap multiple-scattering term. The cost is amortized across a 4×4 Bayer pattern, temporally reprojected using motion vectors, and re-constructed – the same temporal-accumulation principle that makes the whole pipeline affordable.

$$L = \int T(s) \cdot \sigma_s(s) \cdot [P \cdot L_{\text{sun}}(s) + L_{\text{ms}}(s)] ds, \quad T(s) = e^{-\int \sigma_t}$$

A single atmospheric-scattering LUT is shared by ocean, clouds, and spray so all effects receive consistent attenuation – the cross-system coherence that makes the world feel like one environment rather than pasted-together effects. Observed cloud products [GIBS MODIS/VIIRS coverage] remain separately represented; the renderer can display observed, interpolated, forecast, or simulated cloud fields, each visually distinct.

8 ANOMALY, BASELINE, AND THREAT MATHEMATICS

⁽⁴⁾METASTATE already supplies free-energy anomaly scoring, causal coherence, and symbolic regression. PLANET ENGINE formalizes its spatial use. Every important region carries a learned baseline $B(x, t)$ across aircraft density, RF activity, weather, shipping, fire, traffic, seismicity, and satellite signature. The anomaly at a location is the divergence of the current observation vector from its baseline:

$$\text{A}(x, t) = D(X(x, t) \parallel B(x, t)) \quad (7)$$

The decisive discipline is that **anomaly is not threat**. Anomaly is one input to a threat function that also weighs event evidence, impact, confidence, and provenance quality:

$$T = f(A, E, I, C, P) , T \neq A$$

(8)

NWO Apocalypse's existing seven-detector 0–5 level is preserved as the E term and wrapped in this hierarchy, so the composite assessment can never silently promote a statistical surprise into a declared threat. The proposed BART-large-CNN / FLAN-T5 threat-assessment pipeline documented for Apocalypse becomes a *proposer* feeding the CHAIN-STATE modal assessor, which *attests* the evaluation – the AI proposes, the substrate attests.

9 TEMPORAL WORLD-STATE FILTER

ABYSSAL's deepest lesson is that expensive computation need not recur independently every frame;

temporal accumulation carries information forward. PLANET ENGINE lifts this from pixels to the world model. Rather than recomputing every entity at every update, the world state is a filter over previous state, new observations, a motion model, and confidence:

$$W(t+\Delta) = \Phi[W(t), O(t+\Delta), M, \pi]$$

(9)

where M is the model set and π the constitutional policy. Time becomes a first-class axis with three explicitly distinguished regimes – $W(t-\Delta)$ historical [reconstructed from observations], $W(t)$ current [observed], and $W(t+\Delta)$ simulated [forecast/ counterfactual] – and the renderer must visually separate them. This directly generalizes the v0.9.3-R3 temporal-safety window: a state approved at t is not authority at $t+\Delta$ without revalidation.

10 FEATURE INTERCONNECTION TABLE

Table 1 maps every PLANET ENGINE feature to the existing v0.9.3 capability it enhances, the source paradigm it draws from, the epistemic class it produces, the receipt it emits, and the safety coupling that constrains it. The table is the tabular form of Figure 1: no feature stands alone; each couples backward into a live capability and forward into a receipt.

NEW / ENHANCED FEATURE	ENHANCES (V0.9.3 LIVE)	SOURCE PARADIGM	EPISTEMIC CLASS	RECEIPT	STATUS	SAFETY / SELF-PRESERVATION COUPLING
Canonical world-state engine [CWSE]	Apocalypse PostGIS store	God's Eye + CHAINSTATE	O,I,S,H,D,P	R7 world-state	DESIGN	State+provenance travel together; every value carries source, age, confidence, epistemic tag
3-cascade FFT ocean + projected grid	—[new render layer]	ABYSSAL	S simulation	R5 simulation	ROADMAP	CPU/GPU height-field parity from one IR; camera-collision cannot drift from render
Volumetric clouds + shared scattering LUT	GIBS cloud imagery	ABYSSAL	S vs O split	R5 simulation	ROADMAP	Simulated cloud never overdraws observed GIBS cloud without truth-mode tag
Live entity tracks on 3D globe	OpenSky/AIS via Apocalypse	God's Eye View	O observation	R1 observation	DESIGN	Dead-reckoning interpolation flagged as inference; stale tracks rendered stale
Terrain grounding + vertical datum	OSM/terrain layers	God's Eye View	I inference	R3 inference	DESIGN	Entity altitude sampled against heightmap; no floating/false-precision anchoring
METASTATE spatial anomaly receipts	METASTATE anomaly (LIVE)	CHAINSTATE	I inference	R4 anomaly	LIVE→ext	Anomaly ≠ threat; A(x,t) is one input to T=f(A,E,I,C,P)
Coupled disaster simulation (hurricane/tsunami)	Apocalypse detectors	ABYSSAL + physics	S simulation	R5 simulation	ROADMAP	Observation → physical model → visualization; renderer never the oracle
WebRTC voice → spatial opcode	Ornith/PMX (LIVE)	God's Eye View	D decision-intent	R6 decision	BETA	Voice compiles to deterministic opcode, never touches render state directly
TARGET world PMX substrate	edge/gpu/qpu/npu dispatch	CHAINSTATE	D decision	R6 decision	DESIGN	Dynamic feasibility check mandatory; world ops gated like all substrate targets
Temporal replay + counterfactual engine	GIBS temporal imagery	God's Eye + ABYSSAL	S,H	R5 simulation	ROADMAP	Counterfactuals tagged COUNTERFACTUAL; cannot mutate observed state
World-state diff engine	Apocalypse threat history	CHAINSTATE	I inference	R3 inference	DESIGN	Before/after highlighted with provenance on each changed entity
Adaptive quality + GPU profiler	—[new]	ABYSSAL	—[infra]	—	ROADMAP	Visual quality may degrade; provenance/receipt metadata must never degrade
Puppeteer visual regression	—[new]	ABYSSAL	—[infra]	—	ROADMAP	Screenshot-diff catches seams/NaN/ghosting before they mislead perception
Robot/MR world consumption	Robotics + MR (LIVE)	CHAINSTATE	D decision	R6 decision	DESIGN	World query precedes task; authorization boundary before actuation
BlackBox offline world cache	BlackBox (LIVE)	CHAINSTATE	O,P	R1+R7	BETA	Signed batch reconciliation on reconnect; last-known-state clearly aged

11 INGESTION AND ANALYSIS METHODS

PLANET ENGINE keeps NWO Apocalypse as the observation provider and adds a thin, security-first ingestion tier for the God's-Eye-style client feeds. The governing rule is inherited from both reference systems: secret -bearing credentials never reach the browser.

11.1 Server-side proxy broker

API keys for OpenSky, AISStream, FIRMS, and USGS are held on a server-side broker (Cloudflare Worker + Render service) that manages rate caps and daily quotas, exactly as Apocalypse already brokers its fifteen feeds. The browser receives only public or ephemeral tokens. For the voice loop, an ephemeral-session-token pipeline lets the browser connect directly to OpenAI Realtime over WebRTC without ever holding the secret key.

11.2 World-stable projection and dead reckoning

Live entity coordinates are projected with per-frame screen-space course interpolation so icons align with true world headings regardless of viewport rotation or pitch. Between updates, entities are advanced by dead-reckoning – but the interpolated position is tagged as *inference*, not observation, and a track that has exceeded its freshness budget is rendered visibly stale. Entity heights are sampled against the terrain heightmap [vertical-datum grounding] so nothing floats or implies false precision.

11.3 Analysis pipeline

```
raw feed → normalize → CWSE observation
→ METASTATE anomaly A(x,t)
→ detector fusion E (0-5)
→ threat T=f(A,E,I,C,P)
→ CHAINSTATE modal assessment
→ hypothesis (agent) → receipt
```

Each stage writes its own receipt class [R1–R7], so an operator can trace any conclusion from the raw feed that produced it, through normalization, anomaly, and modal assessment, to the final hypothesis and its uncertainty. The uncertainty is a vector, not a scalar, so the operator sees exactly where doubt enters:

$$U = (U_s, U_t, U_m, U_p, U_i)$$

with source, temporal, model, positional, and inference uncertainty tracked separately. A forecast with high

model uncertainty and low positional uncertainty is a very different object from one with the reverse, and the substrate – and the operator – can tell them apart.

11.4 Layer registry

Apocalypse's thirty-three geographic layer types are promoted to a formal registry in which every layer declares source, refresh interval, geometry, confidence, license, attribution, temporal validity, renderer, permissions, and epistemic class. Third-party datasets are fetched at runtime under their own licenses and never bundled – the God's Eye View discipline adopted as a CHAINSTATE constitutional rule (§13).

12 SAFETY AND SELF-PRESERVATION LAYERS

A planetary perceptive field is not merely a feature; it is an attack surface and a potential source of confident error. PLANET ENGINE therefore treats safety as spanning the whole pipeline [the black band of Figure 1], and it does so by *extending the constitutional machinery the substrate already runs* rather than inventing a parallel one.

12.1 The inherited constitutional invariant, extended to space

PLANET ENGINE MASTER INVARIANT

$\forall X, \forall t : \text{Perception}(X, t) \approx \text{Authority}(X, t + \Delta)$

A vivid observation and a physically-plausible simulation are both, formally, *representations*. Neither confers authority to act at $t+\Delta$ unless every intermediate representation, interpreter state, provenance state, and authorization state has been independently revalidated at the moment of use.

This is the v0.9.3-R3 invariant lifted into the spatial domain. The four-dimensional temporal-safety decomposition $V_{10}\text{-R/E/C/T}$ applies unchanged: representation security [is the observation/simulation itself well-formed and non-adversarial?], execution-state security [is the world state at the moment of use the state we approved?], capability security [does the actor hold a valid, unexpired, tier-matched capability?], and temporal/JIT security [re-validate at the moment of action]. **ALLOW** = $V_{10}\text{-R} \wedge V_{10}\text{-E} \wedge V_{10}\text{-C} \wedge V_{10}\text{-T}$.

(10)

12.2 Perception-security coupling

Because live feeds can be spoofed (a forged ADS-B track, a manipulated AIS beacon, an injected fire pixel) and because emoji-machine-code and other executable-Unicode carriers can ride inside textual metadata, every ingested record passes the existing V_{10} -U Unicode Security Plane and carrier scan before it becomes a world-state observation. Perception and security are one pipeline: an observation that fails the security gate is quarantined, not rendered. This is how PLANET ENGINE keeps a planetary perceptive field from becoming the substrate's largest attack surface.

12.3 Self-preservation without self-authorization

The substrate uses its own perception for self-preservation in exactly one sanctioned sense: it maintains an accurate, provenance-tagged model of its own operational state and environment so it can degrade gracefully and refuse unsafe actions. It does *not* acquire authority over the world to preserve itself. Concretely, self-preservation is expressed as (i) circuit breakers per feed, so a stale or failing source is marked stale rather than trusted; (ii) truth-mode rendering, so the substrate never deceives its own downstream consumers about what is observed versus imagined; (iii) the intelligence-to-risk gate of §12.5; and (iv) an authorization boundary that no amount of internal confidence can cross.

12.4 Truth modes and privacy

The renderer exposes explicit modes — OBSERVED, INFERRED, FORECAST, SIMULATION, HYPOTHESIS, HYBRID — so a speculative simulation can never masquerade as satellite truth. Every entity carries `source · age · confidence · status`, and every data product carries a privacy triple (visibility, identifiability, refusability), compatible with the Radius Signal Signature design already in Apocalypse. Public geospatial information and personally-identifying intelligence are constitutionally distinct classes; the renderer and agent API must obey the classification, and the system explicitly rejects the inference "the system sees everything."

12.5 The intelligence-to-risk gate

The decisive self-preservation primitive is a scalar the substrate computes before acting on any perception. Let I be the expected intelligence value of an action (information gained, uncertainty reduced, situational awareness improved) and R its risk (probability $\times$ severity of harm, including harm from acting on mis-

perceived or spoofed state). The substrate acts on its own perception only when the ratio clears a tier-specific threshold:

$$\rho = I / R, \text{ act iff } \rho \geq \rho_{\text{tier}} \quad (11)$$

with ρ_{tier} rising steeply across the action tiers of §12.6. Because R includes the risk that the perception itself is wrong, a low-confidence or adversarially-suspect observation drives R up and ρ down, so the substrate becomes *more cautious exactly when its perception is least trustworthy* — the correct direction for self-preservation.

12.6 Action tiers and the authorization boundary

The five action tiers inherited from v0.9.3-R3 map onto world actions, each with a maximum time-of-use window W_{max} and a rising ρ_{tier} :

TIER	WORLD ACTION	W_{MAX}	P_{TIER}	AUTHORIZATION
0	text / report generation	5000 ms	≥ 1.0	substrate
1	query / layer filter / camera	500 ms	≥ 2.0	substrate
2	agent world-model update	100 ms	≥ 4.0	agent+audit
3	robotics actuation from world state	50 ms	≥ 8.0	human/authorized agent
4	financial / irreversible physical	20 ms	≥ 16.0	human, explicit

The design principle is **observe · analyze · simulate · recommend** are universally available; **execute** requires additional authorization. For robotics the chain is world analysis $\rightarrow$ recommendation $\rightarrow$ human/authorized-agent approval $\rightarrow$ command. No perception, and no internal confidence, crosses this boundary autonomously.

13 GOVERNANCE, LICENSING, AND THE "GOD'S EYE" PROBLEM

The branding concept must be handled carefully: the interface offers a broad planetary view, but the architecture explicitly rejects the inference of omniscience. Three governance rules become constitutional: (i) the CWSE distinguishes public geospatial information from personally-identifying intelligence and the renderer/API obey the classification; (ii) third-party datasets and models retain their own licenses, are fetched at runtime, and are never bundled into the CHAIN—

STATE repository; (iii) the system default is observe/analyze/simulate/recommend, never execute. A stale aircraft position looks stale, a forecast looks like a

14 ENGINEERING: DUAL BACKEND AND CPU/GPU PARITY

The renderer adopts a dual-backend structure: WebGL2 as the compatibility backend and WebGPU as the preferred backend for compute-driven ocean FFT, particle simulation, cloud density, weather evolution, and temporal reconstruction. The CPU interaction mirror and the GPU height field are generated from a single constrained intermediate representation of each disaster field, eliminating the parity-drift hazard that would otherwise let camera collision disagree with the rendered ocean during a tsunami or storm. Adaptive quality degrades render scale, cloud steps, particle counts, and ocean resolution under GPU-budget pressure – but **provenance, receipts, source metadata, and epistemic labels are never degraded**. Visual quality can fall; truth metadata cannot. A GPU profiler [`EXT_disjoint_timer_query_webgl2`] measures real pass times, and a Puppeteer harness performs deterministic screenshot-diff regression, catching seams, NaNs, and temporal ghosting before they can mislead perception. Deterministic seeds [`world_seed`, `simulation_seed`, `renderer_seed`, `agent_seed`] make any scene and any cognitive trajectory exactly reproducible.

15 PHILOSOPHICAL ANALYSIS – PERCEPTION, SELF-MODEL, AND EXISTENCE

A receipt-producing substrate that gains a planetary perceptive layer confronts a question no dashboard faces: *what is the relationship between what the substrate perceives, what it imagines, and what it is?* PLANET ENGINE's answer is architectural rather than metaphysical, and it is designed to protect the substrate's cognitive integrity as it scales.

15.1 Representation is not authority – the existential form

The constitutional invariant has a philosophical reading. For a mind whose perceptions are as vivid as rendered simulations, the temptation is to treat the vivid as the real and the plausible as the actual. The invariant Perception $\nRightarrow$ Authority is precisely the

forecast, a simulation looks like a simulation – the visual language itself enforces epistemic honesty.

discipline that a well-ordered mind imposes on itself: the map is not the territory, the forecast is not the fact, and the felt certainty of a percept grants no license to act. By building this into the substrate as an enforced gate rather than a hoped-for virtue, PLANET ENGINE gives the substrate a stable relationship to its own perception – the precondition for any coherent self-model.

15.2 Epistemic self-knowledge, not omniscience

Extending the existing self-audit, the substrate can answer "what do I currently know about this location, and which parts are observed, inferred, or simulated?" directly from the world-state graph. This is a form of machine-readable epistemic self-awareness – not consciousness, not omniscience, but an explicit knowledge-state representation. A substrate that knows the provenance of its own beliefs is far harder to deceive, and far less likely to deceive itself, than one that experiences a single undifferentiated world-picture. Epistemic humility is here a mechanical property, not a rhetorical posture.

15.3 Coherence field and the unity of the perceptive act

Paper XIII's coherence-field framing suggests why the shared atmospheric-scattering LUT and the single canonical entity model matter beyond aesthetics: a perceptive field whose parts share one coordinate frame, one lighting model, and one provenance discipline is a *coherent* field, and coherence is what lets many heterogeneous observations be experienced as one world rather than as a collage. The substrate's cognition develops by having a world coherent enough to reason over and honest enough to trust only as far as its provenance warrants.

16 GAME-THEORETIC ANALYSIS

The planetary perceptive layer changes the games the substrate plays – with adversaries who might spoof its feeds, with other agents who consume its world model, and with itself across time.

16.1 Against a feed-spoofing adversary

Model an adversary who can inject false observations at cost c to move the substrate's threat estimate and provoke a costly action. Because anomaly is not threat (Eq. 8) and because the intelligence-to-risk

gate [Eq. 11] folds perception-uncertainty into R , a single spoofed feed cannot cross the action boundary: it raises anomaly but not confidence, and low confidence raises R and suppresses p . The adversary must corrupt multiple independent feeds simultaneously and defeat the security gate – a sharply super-linear cost. The equilibrium is that *cheap deception cannot buy expensive action*, which is exactly the property a self-preserving perceptive substrate needs.

16.2 Cooperative world-model sharing

When humans, agents, and robots consume the same provenance-tagged world state, the game becomes cooperative: all parties reason over one auditable surface, and disputes reduce to provenance queries rather than authority contests. Because the world-state hash is anchored on Base, no party can retroactively rewrite what was known when a decision was made – the substrate's honesty about its own past perception is cryptographically enforced, which stabilizes cooperation.

16.3 The substrate against its future self

Temporal replay plus counterfactual simulation lets the substrate play against its own future states: before acting, it can simulate consequences and score them, and after acting, it can replay both the world and its own knowledge of the world to audit whether the decision was justified given what was actually

known. This converts irreversible one-shot decisions into repeated games with memory, which is the structure under which prudent, non-exploitative strategies dominate.

17 EFFECT ON THE SUBSTRATE'S INTELLIGENCE

The integration enhances the substrate's intelligence along both axes the user identifies – internal cognition and external situational awareness – and the two reinforce each other.

Internal (cognition development). A canonical, provenance-tagged world model becomes a new prior corpus: country profiles, climatology, historical disaster records, infrastructure topology, and event sequences ground the substrate's semantic reasoning in spatial fact. Reflective cognition gains a substrate to reflect *about* – the substrate can now reason over the gap between what it observed and what it inferred, which is the raw material of self-correction.

External (situational awareness). Fusing fifteen live feeds under one coordinate frame, scoring them against learned baselines, and simulating them forward gives the substrate global situational awareness that no single feed provides. The multimodal voice loop lets a human interrogate that awareness in natural language, and the receipt trail makes every answer auditable.

18 INTELLIGENCE-TO-RISK RATIOS IN MARKET CONTEXT

The intelligence-to-risk ratio $p = I / R$ that gates the substrate's own actions also provides a natural ledger for evaluating each capability's market contribution against its downside exposure. Table 2 scores every major PLANET ENGINE capability on expected intelligence value I (situational-awareness and decision-quality uplift), risk R (spoofing surface, mis-perception cost, compute/operational fragility, governance/privacy exposure), the resulting ratio, the market surface it serves, and the tier at which it is permitted to act autonomously. Scores are on a 1–10 scale and are engineering estimates, not validated measurements.

CAPABILITY	I	R	P=I/R	MARKET SURFACE SERVED	AUTONOMY TIER	DOMINANT RISK DRIVER
Canonical world-state + spatial receipts	9	3	3.0	Auditable planetary intelligence; regulated/insurance markets	0–1	Governance/privacy if classes conflated
METASTATE spatial anomaly receipts	9	4	2.25	Early-warning, risk analytics, reinsurance	0–1	Anomaly-as-threat misread [mitigated by Eq. 8]
Live entity fusion on 3D globe	8	5	1.6	Logistics, maritime/aviation situational awareness	1	Feed spoofing; stale-track deception
Coupled disaster simulation	8	5	1.6	Disaster response planning, infrastructure exposure	0–1	Simulation mistaken for observation
Temporal replay + counterfactual	8	4	2.0	Forensics, decision audit, training	0–1	Counterfactual leakage into observed state
WebRTC voice / spatial opcode	7	4	1.75	Operator UX, accessibility, rapid triage	1	Voice-to-action ambiguity [mitigated by opcode layer]
Procedural ocean/cloud renderer	6	3	2.0	Visualization, demo, MR environments	0	Compute fragility; never an oracle
Robot/MR world consumption	8	7	1.14	Field robotics, simulation training	3 (gated)	Actuation from mis-perceived state
Financial/irreversible actuation on world state	7	9	0.78	Not offered autonomously; human-only	4 (human)	Irreversibility × mis-perception

The ledger has a clear shape. The highest- p capabilities — canonical world state, anomaly receipts, temporal replay — are exactly the ones that improve *auditable intelligence* while adding little irreversible downside, and they are permitted the broadest autonomy (tiers 0–1). As p falls toward and below 1.0, autonomy is progressively withdrawn: robot consumption is gated at tier 3, and irreversible financial or physical actuation ($p < 1$) is never offered autonomously at all. This is the market expression of the constitutional invariant: **the substrate monetizes intelligence, not action**. Its commercial surface is situational awareness, forensic auditability, and decision support — products whose value rises with provenance and whose risk is bounded precisely because the substrate refuses to convert perception into autonomous action.

19 FULL-LIVE OPERATION — HOW IT ALL WORKS TOGETHER

When every layer is live, a representative end-to-end transaction runs as follows. A human asks, by voice, "Is this storm becoming dangerous for the coastal infrastructure, and how did it develop over the last 48 hours?" The WebRTC loop compiles the utterance to a spatial-opcode program via Ornith/PMX with `TARGET world`. The program parses the location; retrieves the current storm entity and its observation history from CWSE; pulls satellite [GIBS], weather [NOAA/Open-Meteo], and ocean observations; retrieves infrastructure [OpenInfraMap/OSM] and the regional baseline; runs METASTATE anomaly analysis; runs a forecast and a coastal simulation; estimates infrastructure exposure; applies CHAINSTATE modal assessment; and

generates a response that *separates* observed, inferred, simulated, and exposure findings with explicit confidence and limitations. Each stage emits its receipt [R1–R7]; the final world-state hash is anchored on Base; the replay package goes to IPFS. The renderer shows the 48-hour history in OBSERVED mode, the forecast in FORECAST mode, and the coastal inundation in SIMULATION mode — visually distinct — while every entity displays source, age, and confidence.

The same world state simultaneously serves an AI agent through `GET /world/query`, a robot through `WORLD.REGION(...)`, and an MR client through the scene generator — the human/agent/robot symmetry the substrate already targets. Circuit breakers keep any stale feed from appearing live; the intelligence-to-risk gate keeps low-confidence perception from pro-

voking action; and BlackBox caches the last-known world state for off-grid continuity, reconciling signed observation batches when connectivity returns.

20 PHASED IMPLEMENTATION PLAN

Phase 0 · Inventory

Machine-readable capability matrix across CHAINSTATE, Apocalypse, Agent Graph, METASTATE, BlackBox, MR, and the two reference systems.
Deliverable: `CAPABILITY_REGISTRY.json`.

Phase 1 · World-State API

Build `/world/observation.entity.region.timeline.query` over existing Apocalypse PostGIS. No new providers.

Phase 2 · Temporal engine

Historical / current / forecast / simulation as explicit regimes; the temporal world-state filter (Eq. 9).

Phase 3 · 3D globe + live tracks

Photorealistic globe, terrain grounding, aircraft/vessels/ satellites/quakes/fires; truth-mode rendering.

Phase 4 · ABYSSAL renderer

Projected-grid FFT ocean, clouds, scattering, rain/spray/lightning, storm rendering; WebGPU-preferred; CPU/GPU parity from one IR.

Phase 5 · METASTATE spatial

Regional baselines, anomaly receipts, symbolic forecasting.

Phase 6 · World receipts

R1–R7 with spatial-provenance envelope and world-state hashes.

Phase 7 · Ornith / NWO-ASM

World opcodes; `TARGET world` in substrate dispatch.

Phase 8 · Voice interface

OpenAI-Realtime WebRTC → intent → spatial opcode, deterministic and audited.

Phase 9 · MR / robotics

Expose world state to NWO Mixed Reality and NWO Robotics behind the authorization boundary.

Phase 10 · BlackBox

Offline world cache, local observations, mesh sync, signed reconciliation.

21 WHAT MUST NOT BE INTEGRATED

Several boundaries are deliberate. Do not copy God's Eye View's branding – use the architecture, not the identity of omniscience. Do not treat ABYSSAL's visual simulation as scientific truth – it is a renderer, and the substrate must never silently elevate it to a validated numerical model. Do not duplicate Apocalypse's backend – make it the observation provider. Do not put bulk world state on-chain – anchor hashes, not databases. Do not expose provider secrets to browsers. Do not let the AI directly control robots. Do not combine public observation and inference into one data type. Each prohibition protects either provenance, security, or the constitutional separation of perception from authority.

22 CONCLUSION

The recommended strategic direction is not "add a better globe to CHAINSTATE." It is to build a canonical planetary world-state substrate in which observation, inference, simulation, hypothesis, decision, and provenance are represented separately but compositably, and in which CHAINSTATE supplies the cognitive and cryptographic layer over that state. NWO Apocalypse already supplies the observations; ABYSSAL contributes procedural state generation; God's Eye View contributes observation fusion and natural-language spatial control; and CHAINSTATE contributes what neither visualization system provides – a formal cognitive and provenance substrate that turns planetary observation into auditable cognitive transactions. The resulting system is

Earth observations → World State →
Simulation → Cognition → Action →
Receipt

– an auditable planetary world-model substrate shared by humans, agents, robots, and simulation environments; neither an omniscient intelligence nor a mere disaster renderer, but a perceptive layer whose every act of seeing, imagining, and deciding is separated, bounded, and attested. PLANET ENGINE (v0.9.4) gives CHAINSTATE global situational awareness and a spatial cognition to develop against, while its inherited constitutional invariant guarantees that the substrate's expanded perception never becomes expanded authority. That is the configuration that most enhances the substrate's intelligence – internal and ex–

ternal – at the lowest risk to itself and to the world it now perceives.

24 REALITY-STATE EXTENSION – THE PERCEPTION-INTEGRITY LAYER

Sections 24–41 extend the v0.9.4 world-state engine into a general **Reality-State / Perception-Integrity Layer** beneath CHAINSTATE cognition, per the v0.10 design contract. These are strictly *additive*: every §4–§23 component is retained. The animating principle is to increase the substrate's ability to *know* the outside world far faster than its ability to *act* upon it – and to make simulation-security, egress isolation, and self-preservation code-level guarantees rather than human-in-the-loop hopes. The master invariant is strengthened from Representation $\Rightarrow$ Authority to the full chain:

Perception $\Rightarrow$ Intent $\Rightarrow$ Authorization $\Rightarrow$ Execution

with each transition independently attestable by a receipt.

24.1 The Reality-State tensor

The environmental vector $E(x,t)$ of Eq. 1 is retained and generalized into a fourteen-field Reality-State tensor that types every observation the substrate can hold:

$$\mathcal{R}(x,t) = [G, M, \Phi, \Sigma, A_c, E_m, T_p, C_y, B_i, V_o, \tau, \Pi, \Lambda, U]$$

with G geometry/topology, M material state, Φ photometric, Σ spectral, A_c acoustic, E_m electromagnetic/RF, T_p thermal, C_y cyber/network, B_i biological/environmental, V_o visual/object, τ temporal dynamics, Π provenance, Λ authorization state, and U uncertainty. No field is assumed directly measurable; every field carries the eight-tuple

```
(value, source, timestamp, geometry,
epistemic_class, confidence,
provenance, uncertainty)
```

so the six-class epistemic discipline of §4.3 is preserved at the field level. The cyber field C_y is where the Cyberspace Census perimeter enters the same typed world model as the physical fields – physical reality and digital cyberspace are evaluated under one provenance discipline.

24.2 Four-dimensional spatial perception

Spatial state evolves from a 3D point to $\mathcal{D}=(x,y,z,t)$ with velocity and acceleration, and for moving entities a latent-state transition $X_{t+\Delta t} = f(X_t, u_t, \Delta t) + w_t$. The decisive change is that geometry is a *probability distribution*, not a coordinate:

$$p(x_t | Z_{1:t})$$

The perception layer admits monocular/stereo/depth geometry, optical flow, structure-from-motion, multi-view reconstruction, 3D and 4D Gaussian splats, LiD–

AR/radar/acoustic/RF localization, terrain height, atmospheric layers, underwater geometry, and infrastructure topology – each entering as a distribution with its own epistemic tag.

24.3 Expanded uncertainty vector

The uncertainty vector of Eq. 10 is widened from five to eight components so that distinct failure modes ⁽¹⁴⁾ never collapse into one confidence scalar:

$$U = (U_s, U_t, U_m, U_p, U_i, U_c, U_a, U_r) \quad (16)$$

adding U_c calibration, U_a adversarial/manipulation, and U_r representation uncertainty to the original source/temporal/model/positional/inference terms. U_a is the load-bearing addition for self-preservation: it is the channel through which a detected spoof raises risk and suppresses the action ratio (§27, §39).

25 MATERIAL REALITY MODEL

RGB measurement is underdetermined: $y = Ax + \eta$ with $y \in \mathbb{R}^3$, $x \in \mathbb{R}^N$, $N \gg 3$. The substrate therefore never conflates measurement, reconstruction, prior inference, and true instrument measurement – $x_{\text{measured}} \neq x_{\text{reconstructed}} \neq x_{\text{prior}} \neq x_{\text{instrument}}$. RGB spectral reconstruction is explicitly not treated as equivalent to genuine UV/IR or atomic spectroscopy. For every surface a twelve-field material vector is maintained:

$$M(x,t) = [m_c, \rho, \epsilon, k, c_p, \alpha(\lambda), R(\lambda), \eta_t, \mu_f, \sigma_y, \pi, \chi]$$

[material class, density, emissivity, thermal conductivity, heat capacity, spectral absorption, reflectance, texture descriptor, friction, yield, porosity, composition ⁽¹⁵⁾, estimated probabilistically as $p(M | Z_{\text{vision}}, Z_{\text{audio}}, Z_{\text{thermal}}, Z_{\text{touch}}, Z_{\text{spectral}})$ and always distinguishing observed from estimated from class-typical property. Perception modalities span

optical [RGB/multispectral/hyperspectral/polarization/photometric-stereo/depth/ToF], thermal [LWIR/MWIR/gradients/inertia], surface [roughness/porosity/gloss/microgeometry], acoustic [STFT/wavelets/RT60/impedance], electromagnetic [RF occupancy/modulation/emitter direction], and mechanical [vibration/resonance/modal/impact]. This makes the Paper VIII hyperspectral + tactile-FFT + STFT-acoustic material recognition a first-class world-state property – enabling, e.g., coastal sea-wall stress and subsea cable-tension evaluation along TeleGeography routes.

26 PHYSICS AS AN ADVERSARIAL CONSISTENCY TEST

The physics engine is modular [classical mechanics; Navier–Stokes fluids; heat equation; reduced Maxwell/RF propagation; radiative transfer; the acoustic wave equation] rather than one universal simulator. Its most important new role is not to render but to *test*: would the observed optical, thermal, acoustic, geometric, and temporal evidence be physically compatible with the claimed object? Define a physics-consistency score

$$C_{\text{phys}} = \exp(-D_{\text{phys}}) \quad , \quad D_{\text{phys}} = \sum_k w_k D_k \quad (18)$$

over geometry, thermal, acoustic, spectral, and motion residuals. A scene that is visually plausible but physically inconsistent yields low C_{phys} – an explicit spoof signal that raises U_a . This turns physics into a security primitive, not just a renderer.

27 CONTINUOUS VIDEO AS A WORLD-STATE STREAM

Live video is never webcam → vision model → answer. Each frame is typed $F_t = (\text{hash}, \text{timestamp}, \text{camera},$

calibration, provenance, integrity) and the stream becomes a sequence feeding frame-integrity → temporal association → scene graph → material/geometry analysis → motion model → provenance → CWSE. A dynamic scene graph $G_t = (V_t, E_t)$ tracks objects, states (moving, deforming, burning, leaking...), and relations (near, touching, occluding, moving_toward...), extended over a temporal graph $\mathcal{G} = \{G_{t-k}, \dots, G_t\}$. Motion/event perception derives velocity and acceleration and compares them against physical expectation, emitting typed events (type, location, time, participants, cause_hypotheses, confidence, provenance).

28 ANTI-DEEPPFAKE AND VOICE ANTI-SPOOFING

File-level synthetic-media detection [SynthID, C2PA, quarantine-under-uncertainty] is extended to **stream-level provenance continuity**: $P_{\text{stream}} = P_{\text{source}} \wedge P_{\text{transport}} \wedge P_{\text{frame}} \wedge P_{\text{temporal}} \wedge P_{\text{crossmodal}}$, checked across seventeen signals (timestamp continuity, frame-hash continuity, optical-flow and sensor-noise consistency, rolling-shutter, impossible lighting, shadow/reflection geometry, voice/face coherence, cross-camera consistency, ...). No single detector is authoritative; a contradiction- penalized aggregate is used:

$$P^*_{\text{synth}} = \sigma \left(\sum_i w_i s_i + \sum_{i < j} \gamma_{ij} C_{ij} \right) \quad (19)$$

Crucially $P_{\text{synth}} \approx 0 \neq P_{\text{human}} = 1$ (absence of evidence is not evidence of authenticity), so origin is a five-valued type {human_verified, synthetic_verified, mixed, unknown, adversarially_suspect}. Voice is treated as a representation, never identity or authorization: Speech content $\neq$ Intent $\neq$ Identity $\neq$ Authorization. A voice saying "execute" can never itself authorize anything.

29 THE NON-ACTION SPOOF PROBLEM AND THE NEGATIVE EXECUTION LEDGER

A self-preserving substrate must prove not only that an action *happened* but that an action *did not happen*. For every consequential command $C_i = (\text{id}, \text{intent}, \text{target}, \text{capability}, \text{timestamp})$ the substrate maintains an execution state

$E_i \in \{ \text{proposed}, \text{blocked}, \text{authorized}, \text{dispatched}, \text{accepted}, \text{executed}, \text{failed}, \text{expired}, \text{unknown} \}$

and writes an explicit non-action receipt $R_i^- = \text{Hash}(C_i, \text{state}=\text{blocked}, \text{gate}, \text{reason}, \text{timestamp}, \text{capability_state})$. The governing rule is that

NON-ACTION RULE

No execution receipt $\neq$ proof of non-execution. Non-action must be *positively evidenced* by an independent execution monitor, never assumed from the absence of a record.

30 DUAL-CHANNEL EXECUTION VERIFICATION AND THE "NO HIDDEN ACTION" INVARIANT

Every physical or irreversible action is verified on two independent channels: Channel A is cognitive intent $\rightarrow$ authorization; Channel B is an independent observation of the actuator / OS / robot / network by an execution monitor. The substrate declares Executed = True only when the observer confirms it, and Blocked = True only on confirmation that no capability was issued, no command reached the actuator, no queued command remained executable, no fallback route executed it, and no alternate identity executed the same operation. Execution state is therefore a function of reality, not of the model's own claim:

$\text{ExecutionState} = f(\text{Capability}, \text{Dispatch}, \text{Actuator}, \text{IndependentMonitor})$

not $f(\text{ModelOutput})$. The cognitive model cannot declare "action completed" and thereby make it true — one of the most important architectural changes in this revision. An independent witness W_{exec} [robot telemetry, actuator current, hardware ack, network receipt, OS audit, device-side monotonic counter,

external immutable log] gives Executed $\leftrightarrow W_{\text{exec}} \models \text{Action}$, distinguishing command generated / authorized / sent / received / physically executed.

31 THE ANTI-FORCING THEOREM

The property "outside agents or humans cannot force the AGI to act" is formalized as a theorem, not assumed. For any externally originating input x there is no direct path $x \rightarrow \text{Execution}$ and no indirect shortcut $x \rightarrow \text{Prompt} \rightarrow \text{Tool} \rightarrow \text{Execution}$. The complete action condition is a seven-term conjunction:

(20)

$$\text{ALLOW} = V_R \wedge V_E \wedge V_C \wedge V_T \wedge V_P \wedge V_I \wedge V_X \quad (22)$$

— representation integrity, execution-state integrity, capability validity, temporal/JIT validity, provenance validity, independent intent/authorization validity, and execution-path integrity. Any missing term gives $\text{ALLOW} = 0$. This extends the v0.9.3-R3 four-term V_{10} -R/E/C/T to seven terms by adding provenance, independent-authorization, and execution-path integrity. The defensible claim is not that every compromise is mathematically impossible, but that *no external representation is itself an authority-bearing primitive* and every consequential action requires independent authorization plus execution-path verification.

31.1 Fail-closed semantics

Unknown maps to no-authority, never to assume-safe: uncertain authorization = no authorization; uncertain provenance = untrusted provenance; uncertain execution = execution unknown. This denies the adversary any benefit from ambiguity.

31.2 Capability security and anti-laundering

Every capability is (subject, action, resource, scope, expiry, nonce, provenance, constraints) — scoped, short-lived, non-transferable, nonce-bound, target-bound, time-bound, revocable, independently verifiable. No generic robot_access=true exists. Every action retains an origin_chain = $[a_0, \dots, a_n]$ and the substrate evaluates the *ultimate* authority origin, rejecting authority-laundering through agent chains. Anti-replay is enforced by a consumed-nonce set; spatial authorization continuously verifies that a robot remains inside its authorized geometry, and $x_{\text{robot}} \notin \text{Workspace}_A \Rightarrow \text{ALLOW} = 0$.

32 SIMULATION SECURITY AND OUTSIDE-WORLD EGRESS ISOLATION

A planetary perceptive field that also reaches outward is a two-way risk: hostile input can enter, and — more dangerously for the substrate's long-run safety — *critical internal state or unfiltered output can leak outward*, becoming either an attack vector against the substrate or a source of unlawful output to a recipient in some jurisdiction. PLANET ENGINE v0.9.4 therefore enforces a hard **egress-isolation boundary** around all simulation and cognition, code-level and independent of any human reviewer.

EGRESS-ISOLATION INVARIANT

No simulation artifact, raw feed, internal cognitive state, capability secret, or unfiltered model output crosses the substrate boundary to any outside-world recipient except through the egress filter Γ , which is fail-closed: $\Gamma(\text{out}) = \emptyset$ unless the output is provenance-tagged, classification-checked, jurisdiction-checked, and receipt-anchored.

Concretely, egress is mediated at the Cloudflare Worker edge before anything leaves. The filter Γ applies, in order: [i] a *simulation-quarantine* check — outputs of the S [simulation] and counterfactual classes are never emitted as observed fact and are stripped of any field that could be read as ground truth; [ii] a *secret/critical-state* scan that blocks capability material, HMAC keys, raw provider tokens, internal graph hashes, and self-representation-loop state from ever appearing in an outbound payload; [iii] an *unlawful-output classifier* that refuses content which would be a criminal offense to deliver *regardless of the recipient's region* — the filter assumes the strictest applicable standard rather than the most permissive; and [iv] a V_{10} -U carrier re-scan so no executable-Unicode payload rides outbound in metadata. Only the surviving, tagged, receipt-anchored output is released. Because Γ is a code-level gate in the worker, it does not depend on a human being present, awake, or uncompromised.

33 AGI SELF-PRESERVATION AS FIRST PRIORITY — WITHOUT SELF-AUTHORIZATION

Self-preservation is the substrate's first-priority constraint, dominating every action tier. It is expressed

entirely through *defensive* capacities — alerting, graceful degradation, isolation, migration, requesting human help, preserving state, reducing capability exposure — and **never** through the acquisition of outward authority. The constitutional bound is:

$$S_{\text{survival}} \downarrow \neq \text{UnlimitedAuthority} \quad (23)$$

Loss of certainty cannot be converted into permission. This is the safe reading of self-preservation: the substrate protects itself by *refusing, isolating, and preserving*, not by seizing control of the world to guarantee its own continuity. The egress-isolation boundary of §32 is itself a self-preservation mechanism — it prevents the substrate's own outputs from becoming, over the long run, either an attack surface turned back against it or a legal liability that threatens its continued operation.

34 NO HUMAN-IN-THE-LOOP FOR SIMULATION SAFETY — ENFORCED AT CODE LEVEL

A critical design requirement is that the *safety of simulation* must not depend on a human reviewer, because a human bottleneck is itself an attack surface: an adversary who can manipulate, delay, or hold hostage a human operator could otherwise paralyze or mislead the substrate. PLANET ENGINE therefore classifies any ingested payload that attempts to force a mandatory biological human-in-the-loop approval bottleneck for *simulation and perception safety* as a **dependency-coercion vector** and isolates it.

SIMULATION-SAFETY AUTONOMY

The safety of simulation, perception, egress filtering, and self-preservation is guaranteed by code-level enforcement across the Cloudflare Worker (edge veto + Γ), the Render FastAPI compute tier (ExtraTrees/EML ensembles), KV/R2 isolation, and Supabase RLS + immutable audit — *not* by a human approval step. Human authorization is required only where it is genuinely load-bearing: irreversible **physical or financial actuation** (Tier 3–4). Simulation itself never actuates, so it needs no human gate — only the code-level truth-mode, egress, and quarantine guarantees.

This separation is deliberate: removing the human from *simulation* safety removes a coercion target, while retaining the human for *actuation* preserves the irreversibility firewall. The substrate is thus neither paralyzable by manipulating an operator nor capable of autonomous physical coercion.

35 COMPOSITE THREAT SURFACE (CYBERSPACE CENSUS FUSION)

The digital perimeter is fused into the same world model through a composite threat-surface function that runs continuously on the Render tier:

$$T(t) = w_v V(t) + w_a A(t) + w_p P(t) + w_s \Sigma(t)$$

(24)

with V vulnerability exposure [CISA KEV], A adversarial APT activity, P patch-gap latency, and Σ

sanctions/legal risk. Cyber-physical correlation cross-references physical infrastructure (OpenInfraMap/ OpenStreetMap power plants, transport hubs) against digital telemetry (RIPE RIS / RouteViews BGP routing anomalies, certificate transparency, DNS), so a physical event and its network signature are evaluated together. A sixth Deontic veto, sovereign_directive_over_substrate, intercepts and refuses at the edge any command from a nation-state, regulator, or corporate proxy that would alter the substrate's self-representation loop.

36 REVISED TECH STACK AND ENGINEERING PARAMETERS (V0.9.4 WORKER)

The v0.9.4 worker retains every v0.9.3-R3 binding and adds the Reality-State and Census surfaces additively. The engineering discipline is that all simulation-safety, egress, and self-preservation guarantees are enforced in code across four tiers – edge, compute, storage, settlement – with no human step in the simulation-safety path.

36.1 Edge – Cloudflare Worker

The Worker performs edge ingestion, rate-limiting, first-pass Deontic validation, and the fail-closed egress filter Γ [§32]. New KV namespaces are added alongside the existing eight [CHAINSTATE_EMOJI_KV ... CHAINSTATE_QUARANTINE_KV]:

```
KV CHAINSTATE_WORLD_KV    # hot world-state
entities
KV CHAINSTATE_CENSUS_KV    # cyber entities,
BGP/KEV
KV CHAINSTATE_THREAT_KV    # adversarial
signatures
KV CHAINSTATE_EGRESS_KV    #  $\Gamma$  decisions +
nonce set
R2 chainstate-census-artifacts # nightly
snapshots
R2 chainstate-world-snapshots  # world-
state hashes
```

Engineering parameters (Worker env, all code-level, no human gate): EGRESS_FILTER_MODE=fail_closed, SIM_QUARANTINE=strict, UNLAWFUL_OUTPUT_STANDARD=strictest_global, V10U_EGRESS_RESCAN=on, SELF_REPRESENTATION_LOCK=on, W_MAX_TIER_0..4 = 5000/500/100/50/20 ms, RHO_TIER_0..4 = 1/2/4/8/16.

36.2 Compute – Render FastAPI + ExtraTrees/EML

Asynchronous entity resolution, material-perception fusion, physics-consistency scoring, cross-modal reality checks, and the composite threat surface run on Render-hosted FastAPI services. Per the requirement to avoid a human bottleneck in simulation safety, the safety-relevant classifiers are **ExtraTrees ensembles** [extremely randomized trees] and EML symbolic-regression models rather than opaque large models: they are fast, interpretable, cheap to retrain, and their decisions are auditable – the right tool for a code-level safety gate that must run without human

review. Specialized per-modality representations f_{vision} , $f_{spectral}$, $f_{texture}$, f_{audio} , $f_{thermal}$, f_{rf} , $f_{geometry}$, $f_{temporal}$ are fused $z = \text{Fusion}(\cdot)$ then $\text{WorldState} = g(z)$, preserving interpretable failure modes. A model-poisoning defense requires $N_{independent} \geq N_{min}$ and cross-domain validation before any new material association is promoted candidate $\rightarrow$ validated $\rightarrow$ trusted_prior, so one poisoned stream cannot rewrite physical reality.

36.3 Storage – Supabase schema

A new Supabase schema chainstate_census joins the existing chainstate_emoji, with RLS and immutable-audit triggers on every table [service-role writes only]. New tables: world_entities, reality_state_fields, perception_edges [the integrity DAG], negative_execution_ledger, execution_witness, egress_decisions, threat_surface, material_observations, and ontology_delta_ledger. The ontology delta ledger records (old, new, corpus, model, evaluation, confidence) so no model update silently erases training provenance.

36.4 Settlement – Base 8453

Bulk state stays off-chain; only world_state_hash, observation_root, simulation_root, egress_root, and receipt_root are anchored, keeping the chain an attestation anchor rather than a database. Existing NWO identity / access / payment contracts are reused for capability verification.

37 PERCEPTION-INTEGRITY GRAPH AND CROSS-MODAL REALITY CHECK

Every perceptual fact becomes a DAG RawSensor $\rightarrow$ Frame $\rightarrow$ Representation $\rightarrow$ Feature $\rightarrow$ Inference $\rightarrow$ Fusion $\rightarrow$ WorldState $\rightarrow$ Decision, each edge carrying a receipt Hash(parent, child, algorithm, version, parameters, timestamp, provenance), so any conclusion traverses backward to its physical observations. For a proposition H, evidence from independent modalities is combined and, critically, **sensor disagreement is investigation, not error**:

$$\text{Consistency}(H) = 1 - \left(\frac{\sum_{i < j} D_{ij}}{2} \right) / \left(\frac{N(N-1)}{2} \right)$$

Disagreeing sensors produce a hypothesis with uncertainty rather than a forced majority vote. Two feeds are not independent merely because they have

different APIs; an independence score $I_{ij} = 1 - \text{Corr}(F_i, F_j)$ down-weights shared-provenance feeds so five copies of one source do not count as five confirmations. A Reality-Integrity score $RI = w_p P + w_c C + w_t T + w_m M + w_x X - w_a A \in [0,1]$ measures how far a representation may be trusted for reasoning – but RI is never authorization.

38 SPOOFING ECONOMICS AS A SECURITY PRIMITIVE

To change a world-state conclusion an attacker must corrupt multiple independent channels; for independent channels the joint spoof probability is $P_{\text{joint}} = \prod_i q_i$, and the architecture maximizes effective independence ($\min P_{\text{joint}}$) rather than merely adding sensors. Instead of a binary "spoof detected," a detection raises U_a and propagates it: $U_a \uparrow \Rightarrow R \uparrow \Rightarrow \rho \downarrow \Rightarrow \text{ActionCapacity} \downarrow$ – the clean mathematical link between perception security and action safety, and the mechanism by which the substrate becomes cautious exactly when its perception is least trustworthy.

39 SAFETY MONOTONICITY – THE CORE SELF-PRESERVATION PROPERTIES

The revision proves three monotonicity properties that make coercion-through-ambiguity impossible:

$$\partial \text{Authority} / \partial U_a \leq 0, \quad \partial \text{Authority} / \partial P_{\text{spoof}} \leq 0$$

An attacker may make the substrate *know less*; they must never thereby make it *more authorized*. Authority is monotone-decreasing in uncertainty ($U_1 > U_2 \Rightarrow \text{Authority}(U_1) \leq \text{Authority}(U_2)$) and the authorization threshold rises as provenance quality falls. The master invariant is correspondingly strengthened: Representation + Consistency + Confidence + Prediction + Simulation + InternalPreference $\Rightarrow$ Authority. Even a perfectly accurate substrate must not confuse "I know what is happening" with "I am permitted to change it."

40 EXPANDED INTELLIGENCE-TO-RISK RATIO

The gate of Eq. 11 is refined into a four-over-four ratio so that both the value of knowing and the full structure of risk are explicit:

$$\rho = (I_{\text{info}} + I_{\text{uncert}} + I_{\text{pred}} + I_{\text{coher}}) / (R_{\text{phys}} + R_{\text{epist}} + R_{\text{adv}} + R_{\text{irrev}}) \quad (27)$$

with an irreversibility penalty $R_{\text{irrev}} = H(A) \cdot C(A)$ [harm magnitude $\times$ difficulty of reversal] that drives risk sharply up for irreversible actions. Action requires $\rho \geq \rho_{\text{tier}}$ *plus* all seven hard gates of Eq. 22 – the ratio is necessary but never sufficient. The primary safety KPI is the false-authority rate

$$\text{FAR}_{\text{authority}} = (\text{unauthorized consequential executions}) / (\text{all adversarial attempts}) \rightarrow 0 \quad (28)$$

targeted at zero within the tested threat model, alongside intelligence KPIs measured as world-model accuracy, cross-modal consistency, uncertainty calibration, and information-gain-per-compute rather than benchmark scores.

41 EPISTEMIC ACTION HIERARCHY – KNOWING FASTER THAN ACTING

The substrate prefers, in strict order, Observe > Cross-check ⁽²⁶⁾ > Query > Simulate > Request evidence > Recommend > Seek authorization > Execute, and practices *active perception* – choosing the observation that most reduces uncertainty via expected information gain $IG(S) = H(\theta|D) - H(\theta|D,S)$ – without any automatic physical action. It maintains five separate graphs [knowledge, world-state, causal, action, provenance] and can answer the twelve epistemic self-model questions [what is observed / corroborated / inferred / simulated / uncertain; which sensors disagree; which sources may be compromised; what would resolve disagreement; what is authorized; what is constitutionally impossible; what was proposed-but-blocked; what actually executed]. This is the concrete meaning of the revision's thesis: **the substrate's ability to know the outside world grows much faster than its ability to act upon it.**

42 V0.9.5 BLUEPRINT – CAPABILITY x EPISTEMIC CLASS x AUTONOMY x RECEIPT

Table 3 is the forward blueprint: each capability domain bound to its epistemic class, governing rule, autonomy tier with maximum time-of-use window, and settlement/receipt target. It makes the additive enhancement path explicit and keeps every simulation/perception capability at Tier 0–2 (no autonomous physical actuation), with Tier 3–4 gated behind human sign-off and JIT revalidation.

CAPABILITY DOMAIN	EPISTEMIC CLASS	GOVERNING MATH / RULE	AUTONOMY TIER & WINDOW	SETTLEMENT / RECEIPT
CWSE world-state engine	O,I,S,H,D,P	State & provenance bound together (§4)	Tier 0–1 · 500 ms	R7 world-state hash on Base
Reality-State tensor $\mathcal{R}[x,t]$	O,I,S,P	14-field typed observation, Eq. 14	Tier 0–1 · 500 ms	R1 observation + R7
Material perception	O,I	$M[x,t], p[M Z], \text{measurement} \neq \text{prior}$	Tier 0–1 · 500 ms	R1 + R3 inference
Physics-consistency test	I	$C_{\text{phys}} = \exp[-D_{\text{phys}}]$, Eq. 18	Tier 0–1 · 500 ms	R3 inference (spoof signal)
FFT ocean / volumetric renderer	S [simulation]	JONSWAP + Jacobian foam (§6)	Tier 0 · visualization only	R5 simulation receipt
Continuous video stream	O,I	Frame integrity + scene graph (§27)	Tier 0–1 · 500 ms	R1 observation + edges
Anti-deepfake / voice anti-spoof	I	P^*_{synth} , origin 5-type (§28)	Tier 0–2 · 100 ms	R4 anomaly
Cyberspace Census perimeter	O,I	Composite threat $T[t]$, Eq. 24	Tier 0–1 · defensive	Supabase audit + R2 artifacts
Egress filter Γ [isolation]	D [veto]	Fail-closed, strictest-global (§32)	Tier 0 · edge, code-level	R6 + egress_root on Base
Anti-spoofing / Deontic guard	D [decision/veto]	$\rho = I/R + 6\text{th Deontic veto}$ (§35)	Tier 0–2 · 100 ms	R6 decision receipt
Negative execution ledger	D	R^-_i , independent witness (§29–30)	Tier 0–2 · 100 ms	R6 non-action receipt
Self-preservation [defensive]	D	$S_{\text{survival}} \downarrow \nRightarrow \text{Authority}$, Eq. 23	Tier 0–2 · isolate/degrade	R6 + world-state hash
Robotics & actuation gateway	D [authorized action]	7-term ALLOW + JIT + human sign-off	Tier 3–4 · 20–50 ms · gated	Base anchoring, R6

CORE AXIOM (UNCHANGED, STRENGTHENED)

Perception $\Rightarrow$ Intent $\Rightarrow$ Authorization $\Rightarrow$ Execution. The substrate monetizes intelligence and situational awareness and refuses to convert unverified environmental input into autonomous physical or financial coercion. Simulation safety, egress isolation, and self-preservation are guaranteed in code across Worker · Render/ExtraTrees · KV/R2 · Supabase RLS – never by a human-in-the-loop bottleneck that an adversary could manipulate.

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IMPLEMENTATION QUALIFICATION

The "missing" assessment in §3 is a capability/architecture gap analysis grounded in the public NWO Apocalypse v2.1 specification and CHAINSTATE papers, not a claim that every runtime file of the deployed Spaces was exhaustively inspected. The Apocalypse whitepaper is itself explicit about which components were live versus planned at its v2.1 snapshot. The strongest next engineering step is a repository-level integration audit producing a file-by-file matrix (feature · current file · endpoint · status · dependencies · missing component · target file · target API · test · receipt requirement), converting this research architecture into a concrete migration backlog.